

Solve One-Step Addition and Subtraction Equations

Lesson 7-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

$x + 5 = 12$ — both sides equal 12 when $x = 7$

Equation

A math sentence with an equal sign showing both sides are the same.

$$x + 2 = 7$$

balanced with =

Addition undoes subtraction: if $x - 3 = 10$, add 3 to get $x = 13$

Inverse operation

Two math actions that undo each other, like $\times$ and $\div$.

$$x + 2 = 7$$

balanced with =

$x + 5 = 12 \rightarrow$ subtract 5 from both sides $\rightarrow x = 7$ (x is isolated)

Isolate

To get the letter by itself on one side.

$$x + 2 = 7$$

balanced with =

$x = 7$ is the solution to $x + 5 = 12$ because $7 + 5 = 12$
✓

Solution

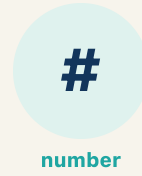
The number that makes the equation true.



$x + 4 = 9$ is solved in one step: subtract 4 from both sides, so $x = 5$.

One-step equation

An equation you can solve in a single step using one inverse operation.



If you subtract 4 from the left side, you must subtract 4 from the right to keep balance.

Balance

Keeping both sides of an equation equal by doing the same thing to each side.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can solve one-step addition and subtraction equations using inverse operations.

1 Notice

Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case.

2 Key idea

I can solve one-step addition and subtraction equations using inverse operations.

3 Words I'll use

Equation

Inverse operation

Isolate

Solution

One-step equation

Balance



Think About It

- What operation connects the locker number and 23?
- What does n represent in the equation?
- How could you undo the addition to find n ?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $x + 9 = 15$

A. $x = 6$

B. $x = 24$

C. $x = 9$

D. $x = 15$

Step 1 Subtract 9 from both sides: $x = 15 - 9 = 6$.

Step 2 Check: $6 + 9 = 15$ ✓

✓ **Answer:** A. $x = 6$

 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $y - 7 = 20$

A. $y = 27$

B. $y = 13$

C. $y = 7$

D. $y = 140$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $n + 15 = 42$

A. $n = 27$

B. $n = 57$

C. $n = 15$

D. $n = 42$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Chen's clue is $n + 23 = 58$, where n is a locker number. Why can't she just 'know' n by looking, and what could she do to both sides to find it?

Level 1 support

Start here: I can undo the addition to get the variable by itself.

 **Need a hint?**

Sentence starters (optional)

The variable n is added to _____, so to find n I can _____.

La variable n está sumada a _____, así que para hallar n puedo _____.

I would use _____ to undo the _____.

Usaría _____ para deshacer la _____.

Word bank: equation inverse operation isolate solution

Listen for: Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n .

Level 2

How would your plan change if the clue were $n - 23 = 58$ instead? Justify which inverse operation you would use.

- For $n - 23 = 58$ I would _____ instead because _____.
- The inverse operation changes because _____.

EXPLORE

What inverse operation did you use to solve $n + 23 = 58$, and why must you do it to BOTH sides of the balance?

Level 1 support

Start here: I do the same inverse operation to both sides to keep it balanced.

 Need a hint?

Sentence starters (optional)

I used ____ to undo ____, so the variable equals ____.

Usé ____ para deshacer ____, así que la variable es igual a ____.

I do it to both sides because ____.

Lo hago en ambos lados porque ____.

Word bank: inverse operation isolate solution equation

Listen for: Students subtract 23 from both sides to get $n = 35$ and explain that doing the same to both sides keeps the equation balanced and the solution true.

Level 2

How can you check that $n = 35$ is the solution? Show why checking matters even when your steps look correct.

- I can check by ____, which gives ____.
- Checking matters because ____.

CONNECT

Where might solving a one-step equation help you find a missing amount in real life?

Level 1 support

Start here: A one-step equation finds an unknown when I know a total and a part.

 **Need a hint?**

Sentence starters (optional)

A one-step equation would help me find ____ when I know ____.

Una ecuación de un paso me ayudaría a hallar ____ cuando sé ____.

I would use the inverse operation ____ to solve it.

Usaría la operación inversa ____ para resolverla.

Word bank: equation inverse operation isolate solution

Listen for: Students give a realistic context (money saved, distance left, missing count) and connect it to isolating a variable using an inverse operation.

Level 2

Critique: 'Adding the same number to both sides changes the answer.' Use the idea of keeping the equation balanced to explain why this is false.

- This is false because ____.
- Doing the same operation to both sides keeps ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words equation, inverse operation, isolate, and solution.

1. Kernel Sentence subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.

Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equation matters in math because ____.
Ecuación importa en matemáticas porque ____.

but
pero

Equation matters in math, but ____.
Ecuación importa en matemáticas, pero ____.

so
entonces

Equation matters in math, so ____.
Ecuación importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The variable n is added to ____ , **so to find n I can** ____ .

La variable n está sumada a ____ , *así que para hallar n puedo* ____ .

I would use ____ **to undo the** ____ .

Usaría ____ *para deshacer la* ____ .

I used ____ **to undo** ____ , **so the variable equals** ____ .

Usé ____ *para deshacer* ____ , *así que la variable es igual a* ____ .

Try It

Solve on your own. Check the answer key when you are done.

1. Solve: $n + 4.5 = 10$.

- A. $n = 5.5$
- B. $n = 14.5$
- C. $n = 6.5$
- D. $n = 45$

Show your work:

2. A savings account had some money. After adding \$25 it has \$90. How much was there before?

- A. \$65
- B. \$115
- C. \$3.6
- D. \$65.5

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Sentence starter: For $x + 25 = 60$, the inverse operation is ____, so $x =$ ____ . Check: ____ . For $y - 25 = 60$, the inverse operation is ____, so $y =$ ____ . Check: ____ . Therefore, ____ .

Show your work:

Reflect — Exit Ticket

Solve: $p + 2.8 = 9.1$

- A. $p = 6.3$
- B. $p = 11.9$
- C. $p = 6.8$
- D. $p = 3.25$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $y = 27$ — *Add 7 to both sides: $y = 20 + 7 = 27$. Check: $27 - 7 = 20$ ✓*
2. **You Try (notes):** A. $n = 27$ — *Subtract 15 from both sides: $n = 42 - 15 = 27$. Check: $27 + 15 = 42$ ✓*
3. **Try It 1:** A. $n = 5.5$ — *Subtract 4.5: $n = 10 - 4.5 = 5.5$.*
4. **Try It 2:** A. $\$65$ — $x + 25 = 90$, so $x = 65$.
5. **Exit Ticket:** A. $p = 6.3$ — *Subtract 2.8 from both sides: $p = 9.1 - 2.8 = 6.3$. Check: $6.3 + 2.8 = 9.1$ ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).